

Redline review of “Remainder Terms of the Riemann Zeta Function” (rewrite v7)

Audit report with numbered proposed changes

Review pass of `main.tex` (4953 lines, August 4, 2026)

August 5, 2026

This report is the requested full review of the paper. **Nothing in `main.tex` has been changed.** Each proposed change is numbered so you can reply with, e.g., “apply 1, 4, 9–12.” Conventions: ~~red-strikethrough~~ is text proposed for deletion, blue is proposed replacement or insertion. Line numbers refer to the current `main.tex`. Tags: **[ERROR]** a mathematical or factual mistake; **[INCONSISTENCY]** two places in the paper that contradict each other; **[UNCLEAR]** correct but likely to confuse a reader; **[TYPO]** grammar, punctuation, citation mechanics; **[EDITORIAL]** housekeeping before submission.

Two of the items below (Changes 1 and 2) were verified numerically before being flagged; the check values are quoted in place.

Part 1 — Mathematical errors and internal inconsistencies

Change 1 **[ERROR]** §9.4 “*PS, AK and RS Legs and Angles,*” paragraph “*Intersecting equal-leg ovals and the Riemann Hypothesis,*” `main.tex` lines 2133–2139

Current:

On the critical line this is automatic: Corollary 9.? puts the whole line in $\mathcal{E}_{\text{ps}}$, and on $\sigma = \frac{1}{2}$ one also has $\Sigma_1 = \Sigma_2$ (up to the usual conjugacy with $|\chi| = 1$), hence $B_{1rs} = \zeta/2$ and the whole line lies in $\mathcal{E}_{\text{rs}}$ as well.

Proposed:

On the critical line this is automatic: Corollary `cor:equal-legs` puts the whole line in $\mathcal{E}_{\text{ps}}$, and the *rs* split is reflection-symmetric there as well: $\Sigma_2 = \chi \overline{\Sigma_1}$ and $R = \chi \overline{R}$ (the latter because $e^{-i\psi/2}R$ is real, Appendix B), so $\chi \overline{B_{1rs}} = \Sigma_2 + \frac{1}{2}R = \zeta - B_{1rs}$; since $|\chi| = 1$ this gives $|B_{1rs}| = |\zeta - B_{1rs}|$, and the whole line lies in $\mathcal{E}_{\text{rs}}$ as well. (Note that $B_{1rs} \neq \zeta/2$: only its perpendicular projection onto the base is $\zeta/2$, by Corollary `cor:bisector-proj.`)

Why: The two struck claims are false. $\Sigma_2 = \chi \overline{\Sigma_1}$, which is not Σ_1 , even on the line; and $B_{1rs} = \Sigma_1 + \frac{1}{2}R$ is not $\zeta/2$ (that would require $\Sigma_1 = \Sigma_2$ literally). Verified numerically at $\sigma = \frac{1}{2}$, $T = 6.18$: $B_{1rs} = 1.9256 + 2.1268i$ while $\zeta/2 = -0.3869 + 1.5494i$; yet the legs are equal, $|B_{1rs}| = |\zeta - B_{1rs}| = 2.868955\dots$, so the *conclusion* (the line lies in $\mathcal{E}_{\text{rs}}$) stands once the argument is repaired as above.

The paper’s own Figure `fig:splits-h` already draws B_{1rs} off $\zeta/2$ ($h_{R/2} = 2.383 \neq 0$), so this passage also contradicts that figure.

Change 2 [INCONSISTENCY] *Positivity of d_1, d_2 : abstract (lines 174–175), §1 item 3 (lines 208–212), §4 (lines 428–443) versus Remark 12.3 `rem:d-ratio` (lines 3452–3458)*

Current:

(§4) Note: both d_1 and d_2 are real and ~~positive~~. ... They are positive because R lies in the cone spanned by the two directions $e^{-i\omega}$ and $e^{i(\omega+\psi)}$... each fractional summand points *forward* along its partial sum, a positive fraction of the next term.

(Abstract) with $\hat{d}_1, \hat{d}_2$ ~~positive real numbers~~, the fractions of those two summands that are used.

Proposed:

(§4) Note: both d_1 and d_2 are real, and generically positive. They are positive whenever R lies in the cone spanned by the two directions $e^{-i\omega}$ and $e^{i(\omega+\psi)}$, so that both numerators carry the same sign as the common denominator $\sin(2\omega + \psi)$. That holds everywhere on the critical line and, off it, everywhere outside a narrow window around each pole of `§sec:pole-locations`; inside such a window R exits the cone and one of the two weights turns negative (Remark `rem:d-ratio`). Geometrically, outside those windows each fractional summand points *forward* along its partial sum, a positive fraction of the next term.

(Abstract) with $\hat{d}_1, \hat{d}_2$ real numbers — positive on the critical line, and off it outside narrow windows around the poles of the split — the fractions of those two summands that are used.

Why: The unqualified positivity claim is contradicted later by the paper itself: Remark 12.3 reports $d_1 = -d_2 \approx -4.07 \times 10^{36}$ near a pole and says explicitly that “ R has left the cone ... in a narrow window around each pole.” Verified numerically: at $\sigma = 0.1$, $T = 4.2566$ one gets $d_1 = -147.3$, $d_2 = +147.7$; at $T = 4.2560$, $d_2 = -1.24$. The same softening is needed in §1 item 3 (see Change 16 for its exact wording). Corollary `cor:equal` and everything on the critical line are unaffected.

Change 3 [ERROR] *Proof of Theorem 4.1, “Integer T ” paragraph, line 504*

Current:

... and the factor ~~$(-1)^{\lfloor t \rfloor}$~~ in R shifts $\phi = \arg R$ by π .

Proposed:

... and the factor $(-1)^{\lfloor T \rfloor}$ in R shifts $\phi = \arg R$ by π .

Why: The sign factor in question is the $(-1)^{\lfloor T \rfloor}$ of the split `eq:Rak` (§7.1); $\lfloor t \rfloor$ (the floor of the height, $t \approx 280$ at $T = 6.18$) changes parity at many non-integer T and plays no role at the handoff instants the paragraph is about. Please confirm the intended convention, but as written the subscripted variable appears to be a typo.

Change 4 [ERROR] *Table 2 caption `tab:poles` (lines 1652–1655) and the same sentence repeated in §9.4 (lines 1998–2000)*

Current:

Note the poles occur at approximately $T = \lfloor T \rfloor \pm \frac{1}{4}$ and approach $\lfloor T \rfloor \pm \frac{1}{4}$ as T approaches ∞ .

Proposed:

Note the poles occur at approximately $T = \lfloor T \rfloor + \frac{1}{4}$ and $T = \lfloor T \rfloor + \frac{3}{4}$ (equivalently, an integer $\pm \frac{1}{4}$), and approach those values as $T \rightarrow \infty$.

Why: Per the table itself, the poles sit near fractional parts 0.25 and 0.75 (e.g. 1.2689 and 1.7637). “ $\lfloor T \rfloor \pm \frac{1}{4}$ ” includes $\lfloor T \rfloor - \frac{1}{4}$, which lies in the *previous* unit interval; the intended reading was presumably “nearest integer $\pm \frac{1}{4}$.” The same wording appears twice (both places listed above), and also matches the correct phrasing already used elsewhere (“near fractional parts ≈ 0.25 and ≈ 0.75 ”).

Change 5 **[ERROR]** §7.3 “Scaling the remainders ...,” equation *eq:kappa* and its lead-in, lines 1092–1110

Current:

The three amplitudes admit the ~~exact formulas~~ ... $\kappa_{1rs} = \frac{1}{4} n^\sigma |I_1(s) + \chi(s)I_2(s)|$...

Proposed:

The three amplitudes admit the closed forms

$$\kappa_{1rs} = \frac{1}{2} n^\sigma |R| \approx \frac{1}{4} n^\sigma |I_1(s) + \chi(s)I_2(s)|,$$

the “ $\approx$ ” holding when R is evaluated by Kuznetsov’s method, with κ_{1ps} and κ_{1ak} as before.

Why: $R_{1rs} = \frac{1}{2}R$ makes $\kappa_{1rs} = \frac{1}{2}n^\sigma|R|$ the exact statement. Writing it as $\frac{1}{4}n^\sigma|I_1 + \chi I_2|$ silently replaces R by R_{ak} , which *eq:Rak* itself marks as “ $\approx$ ”; calling the result an “exact formula” contradicts §7.1. (κ_{1ak} is fine as is: it is the definition of the *ak* split. The second equality in $\kappa_{1ps} = 2\kappa_{1rs} \sin(\cdot)/\sin(\cdot)$ then also becomes exact, with κ_{1rs} read as $\frac{1}{2}n^\sigma|R|$.)

Change 6 **[ERROR]** §11.2 (I_p for L -functions), sentence after *eq:Ip-base*, lines 2563–2565

Current:

Asymptotically $I_p(T) \sim \frac{2p}{(p-1)^2} \pi T^2$, so the leading coefficient is 2π for zeta (~~formally~~ $p=1$), $\frac{3}{2}\pi$ for $p=3$, ...

Proposed:

Asymptotically $I_p(T) \sim \frac{2p}{(p-1)^2} \pi T^2$: the leading coefficient is $\frac{3}{2}\pi$ for $p=3$ and $\frac{5}{8}\pi$ for $p=5$, against 2π for zeta’s own $I(T)$. (Zeta is not the $p=1$ member of this family — the factor $2p/(p-1)^2$ diverges there — because for zeta every summand is a link and there are no phantoms.)

Why: Substituting $p=1$ into $2p/(p-1)^2$ gives a division by zero, not 2π , so “(formally $p=1$)” is not correct as stated. The 2π comes from $I(T) \sim 2\pi T^2$ directly.

Change 7 **[ERROR]** §12.4, equation *eq:S-slope* and the sentence around it, lines 3504–3509

Current:

... and falls smoothly in between, at the rate $S'(t) = -\frac{\theta'(t)}{\pi} \equiv -\frac{1}{2\pi} \log \frac{t}{2\pi}$, which is exactly the mean zero density.

Proposed:

... and falls smoothly in between, at the rate

$$S'(t) = -\frac{\theta'(t)}{\pi} = -\frac{1}{2\pi} \log \frac{t}{2\pi} + O(t^{-2}),$$

which is the mean zero density.

Why: $\theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(t^{-2})$, so the second equality is asymptotic, not exact. This matters here more than elsewhere because §12.4 is explicitly at pains to keep exact and approximate statements separate (“Stated in terms of θ the formula is exact ... approximation enters only when θ is replaced by its expansion”). $S' = -\theta'/\pi$ between ordinates is the exact clause and can keep the “exactly.”

Change 8 **[ERROR]** §2, *description of Siegel’s contour C_2 after eq:siegel13, lines 283–285*

Current:

where the path C_2 is a polyline of two half-lines, one starting at $\eta - \frac{\epsilon}{2}\eta$ and containing $\eta = +\sqrt{\frac{t}{2\pi}}$ and $-(m + \frac{1}{2})$, and g is defined below.

Proposed:

where the path C_2 consists of two half-lines meeting at the point $\eta(1 - \frac{\epsilon}{2})$ for a small fixed $\epsilon > 0$, one half-line passing through the saddle point $\eta = +\sqrt{t/2\pi}$ and the other through $-(m + \frac{1}{2})$, and g is defined below. *[Wording to be confirmed against Siegel §2 / the Barkan–Sklar translation before applying.]*

Why: As written the sentence has three problems: ϵ is never defined anywhere in the paper; only one of the “two half-lines” is described; and “one starting at ... and containing η and $-(m + \frac{1}{2})$ ” cannot be one straight half-line through both points and the stated start. The proposal is the most plausible reconstruction (vertex at $\eta - \frac{\epsilon}{2}\eta = \eta(1 - \frac{\epsilon}{2})$, one arm through each of the two named points), but since this describes Siegel’s construction it should be checked against the source rather than taken from this report.

Part 2 — Unclear passages

Change 9 **[UNCLEAR]** §8.4 “Where are the d_1 and d_2 poles?,” opening of the second paragraph, lines 1611–1615

Current:

A pole occurs when the link at $\lfloor T+1 \rfloor$ of the Σ_1 chain (the summand just after the last summand of the partial sum) is parallel to the link at $\lfloor T+1 \rfloor$ of the Σ_2 chain, that is, when the two next summands have the same argument, ...

Proposed:

A pole occurs when the two bisector links — link $m = \lfloor T \rfloor$ of each chain, carrying summand $\lfloor T \rfloor$, the summand just after the last one included in the partial sum — are parallel, that is, when the two next summands have the same argument, ...

Why: Under the paper’s own numbering (§6: “link n represents the $(n + 1)$ st summand”), the summand $\lfloor T + 1 \rfloor$ is link $m = \lfloor T \rfloor$, not “the link at $\lfloor T + 1 \rfloor$.” The current phrasing quietly switches to summand-numbering for links, which §6 and §9.1 tell the reader not to do; it is also exactly the bisector-link pair, so naming it makes the cross-reference to §9.1 immediate.

Change 10 [UNCLEAR] *Roadmap paragraph, §1, lines 243–263: §8 is missing*

Current:

... §sec:remainders-summary names the three splits R_{rs} , R_{ps} , and R_{ak} side by side, and §sec:remainder-scaling records how the remainders scale when $\{T\}$ is held fixed. §sec:geometry tells the experimental/geometric story ...

Proposed:

... §sec:remainders-summary names the three splits R_{rs} , R_{ps} , and R_{ak} side by side, and §sec:remainder-scaling records how the remainders scale when $\{T\}$ is held fixed. §sec:d1-function studies the weight d_1 as a function of the index: its handoff jumps, two exactly continuous coordinates built from it, and zeta-free closed-form approximations on and off the critical line. §sec:geometry tells the experimental/geometric story ...

Why: The roadmap walks through every section of the paper except §8 (sec:d1-function), which is one of the longer and more technical sections; a reader following the roadmap hits it unannounced.

Change 11 [UNCLEAR] *Draft banner under the title, lines 153–156*

Current:

~~DRAFT, rewrite in progress (v7). Sections 1–8 drafted; §10 and appendices pending.~~

Proposed:

~~DRAFT, rewrite in progress (v7). [or delete the banner]~~

Why: The banner is stale: the paper now has thirteen sections plus two completed appendices. Either update the status or remove the line.

Change 12 [UNCLEAR] §9.4 “The strip lines at ≈ 0.25 and ≈ 0.75 are not flat,” lines 1995–2008

Current:

~~[Pulled from the original paper; to be reworked.] Numerical methods were used to show, relative to σ in the critical strip, where the poles of the split lie. The poles occur at approximately $T = \lfloor T \rfloor \pm \frac{1}{4}$ and approach $\lfloor T \rfloor \pm \frac{1}{4}$ as $T \rightarrow \infty$. Viewed across the strip, the pole heights are nearly invariant (“flat”) in σ , with a symmetry about $\sigma = \frac{1}{2}$: reflecting $\sigma \mapsto 1 - \sigma$ yields the same T -solutions, because $\arg \chi(\sigma + it) = -\arg \chi(1 - \sigma + it)$ by the identity $\chi(s)\chi(1 - s) = 1$ and the pole condition uses $\arg \chi$ only as a phase offset. But they are not exactly flat: ...~~

Proposed:

Viewed across the strip, the pole heights of §[sec:pole-locations](#) are nearly invariant (“flat”) in σ , and symmetric about $\sigma = \frac{1}{2}$ (Table [tab:poles](#) and its caption). But they are *not exactly* flat: there is a slight, symmetric curvature in σ , visible in Figure [fig:pole-heights-zoom](#). . .

Why: Two problems: the bracketed editorial marker “[Pulled from the original paper; to be reworked.]” is still in the text; and the struck sentences repeat the Table 2 caption almost verbatim (the $\pm\frac{1}{4}$ sentence, the $\sigma \mapsto 1 - \sigma$ symmetry argument). The one new fact in this subsection is the curvature; the proposal keeps that and points back rather than repeating. (If Change 4 is applied, apply it here too if this paragraph is kept.)

Change 13 [\[UNCLEAR\]](#) *Remark 12.1* [rem:leg-ratio](#), line 3396: notation collision on ϑ
Current:

... and $\arg \chi = -2\vartheta(t)$, with ϑ the Riemann–Siegel theta function.

Proposed:

... and $\arg \chi = -2\theta(t)$, with θ the Riemann–Siegel theta function of §8.2.

Why: Everywhere else the Riemann–Siegel theta function is $\theta(t)$ (§8.2, §12.4, ...), while ϑ_1, ϑ_2 are reserved for the leg angles defined three pages earlier. Using bare $\vartheta(t)$ here invites the reader to connect it to the leg angles.

Change 14 [\[UNCLEAR\]](#) §7.1, line 909–911: Kuznetsov’s coefficient ω_0 vs. the direction angle $\omega(T)$
Current:

... each evaluated as a short finite series in precomputed coefficients $\omega_0, \omega_1[n], \lambda[n]$; ...

Proposed:

... each evaluated as a short finite series in precomputed coefficients (Kuznetsov’s $\omega_0, \omega_1[n], \lambda[n]$ — his notation, unrelated to the direction angle $\omega(T)$ of §[sec:decomp](#));
...

Why: ω is one of the paper’s most-used symbols; a one-clause disambiguation is cheap insurance.

Change 15 [\[UNCLEAR\]](#) *Abstract, lines 175–177: which equality is verified in Lean*
Current:

As a corollary, when $\sigma = \frac{1}{2}$ one has $\hat{d}_1 = \hat{d}_2$; we have verified this fact in Lean.

Proposed:

As a corollary, when $\sigma = \frac{1}{2}$ one has $d_1 = d_2$ (equivalently $\hat{d}_1 = \hat{d}_2$); the fact is formally verified in Lean.

Why: The Lean file of Appendix B proves the unhatted equality $d_1 = d_2$; the hatted version follows on the line because $|\chi| = 1$ makes the two scaling factors of `eq:frac-vs-weight` equal. Since the abstract introduces only the hatted symbols, either phrasing works, but the parenthetical keeps the claim aligned with what the appendix literally proves.

Change 16 [UNCLEAR] §1 item 3, lines 208–212: d vs. $\hat{d}$

Current:

... each is exactly *one more summand* of its Dirichlet sum, namely the $(m+1)$ st term, scaled by a positive real “fractional” weight ~~d_1 or d_2~~ .

Proposed:

... each is exactly *one more summand* of its Dirichlet sum, namely the $(m+1)$ st term, scaled by a positive real “fractional” weight $\hat{d}_1$ or $\hat{d}_2$ (§`sec:summand` distinguishes these dimensionless *fractional amounts* from the closely related absolute lengths d_1, d_2).

Why: §5 (lines 586–599) is explicit that the coefficient of the $(m+1)$ st summand is the hatted fractional amount, “not the weights d_1, d_2 .” The introduction currently attaches the unhatted symbols to exactly the role the paper later reserves for the hatted ones. (If Change 2 is applied, add its positivity qualifier here as well.)

Part 3 — Typos, grammar, citations

Change 17 [TYPO] §6 opening, line 652–653

Current:

... and denote two complex ~~number~~ by

Proposed:

... and denote two complex numbers by

Why: Singular/plural. (Optionally “define the two split points” since B_1 and B_2 get a geometric name later.)

Change 18 [TYPO] Figure 5 caption (`fig:kuznetsov-zoom`), line 937–938: missing verb

Current:

... Kuznetsov’s approximate half-remainders R_{1ak}, R_{2ak} ~~in orange dashed line~~.

Proposed:

... Kuznetsov’s approximate half-remainders R_{1ak}, R_{2ak} are drawn as the orange dashed pair.

Why: The clause has no verb, and “line” (singular) describes two links.

Change 19 [TYPO] §12.4, equation `eq:B1-rotated`, lines 3546–3549: punctuation

Current:

$B_1 e^{i\theta(t)} = \frac{1}{2}Z(t) + ih,$ $h \in \mathbb{R},$ Multiplying by $e^{i\theta}$ rotates ...

Proposed:

$B_1 e^{i\theta(t)} = \frac{1}{2}Z(t) + ih,$ $h \in \mathbb{R}.$ Multiplying by $e^{i\theta}$ rotates ...

Why: The displayed equation ends with a comma but the next word begins a new sentence (“Multiplying ...”). Change the trailing comma to a period.

Change 20 [TYPO] *Footnote in §9 (before eq:LN), lines 1691–1694* — optional

Current:

... unlike the Dirichlet L -functions of §sec:IT-Lfunctions, ~~who~~ have bounded character sums ...

Proposed:

... unlike the Dirichlet L -functions of §sec:IT-Lfunctions, whose characters have bounded partial sums ...

Why: “Who” is the personal relative pronoun; functions take “which” or “whose.” Flagged as optional because this wording was chosen deliberately in an earlier edit; “which have bounded character sums” is the minimal fix if you want to keep the original rhythm.

Change 21 [TYPO] §11.5 (*sec:yinyang-infinity*), lines 3179–3181: *Pugh cited without a bibliography entry*

Current:

(it appears in §15.3 of Titchmarsh’s book [T] and in §5.4.1 of ~~Glendon Ralph Pugh’s 1998 MS thesis~~).

Proposed:

(it appears in §15.3 of Titchmarsh’s book [T] and in §5.4.1 of Pugh’s thesis [Pugh1998]).
Add to the bibliography: G. R. Pugh, *The Riemann–Siegel formula and large scale computations of the Riemann zeta function*, M.Sc. thesis, University of British Columbia, 1998.

Why: Every other work mentioned in the paper is in the bibliography; Pugh is the one inline-only reference. (Please confirm the thesis title from the copy you used before applying.)

Change 22 [TYPO] §12.2 (*sec:yinyang-links*), lines 3320–3322 — optional

Current:

... it is sequence A005563 in the OEIS database, though no connection of that sequence to the zeta function is listed there.

Proposed:

... it is sequence A005563 in the OEIS [OEIS], though no connection of that sequence to the zeta function is listed there. *Add to the bibliography:* OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A005563 (oeis.org/A005563).

Why: Purely mechanical: OEIS asks to be cited, and the entry gives the reader the link. Low priority.

Part 4 — Editorial / pre-submission housekeeping

Change 23 [EDITORIAL] *Acknowledgements, lines 4859–4866*

Current:

(comment block only) TODO before submission: no Acknowledgments section in this rewrite yet.

Proposed:

Port the Acknowledgements from the v6 folder (`main.tex` line 2439): Chris Cowherd (eight years of visualization and experimental-mathematics tools), Nicolas Fish (three years building and tweaking Zest), Daniel Flores Galiote, Jakob von Raumer, Brian Conrey, Freeman Dyson and family.

Why: You asked to be reminded; the TODO comment is already in the source, but the section itself still needs to be written before submission.

Change 24 [EDITORIAL] *Preamble, lines 96–117: dead macros*

Current:

`\zh`, `\half`, `\Rps`, `\Rtps`, `\Rak` (lines 96–100) and `\figplaceholder` (lines 103–117)

Proposed:

Delete all six definitions.

Why: None of them is used anywhere in the body (checked by `grep`). The placeholder macro was scaffolding for the rewrite; every figure is now real.

Change 25 [EDITORIAL] *§13 title, line 4540 — optional*

Current:

Prior literature ~~on the topic~~

Proposed:

Prior literature

Why: “On the topic” adds nothing; section titles elsewhere in the paper are tighter. Style only — skip freely.

Checks that came back clean

For completeness, things audited that did *not* produce a change request: the proof of Theorem 4.1 (Steps 1–6 recomputed by hand; the integer- T identity $[T]^{iI(T)} = -e^{i\omega}$ is correct, since $I(T) \log(1 + \frac{1}{T}) = \pi(2T+1)$ is an odd multiple of π); the Boole expansion $n^\sigma \Phi(-1, \sigma, n) = \frac{1}{2} + \frac{1}{8n} - \frac{5}{128n^3} + O(n^{-5})$ at $\sigma = \frac{1}{2}$; the follower identity $m(m+2) = (m+1)^2 - 1$ and the factor $[T]^{-1}$; the ordinate counts quoted for the windows $6 \leq T \leq 7$ (55: $\gamma_{117} - \gamma_{171}$), $5.9 \leq T \leq 6.7$ (42: $\gamma_{113} - \gamma_{154}$), and $3 \leq T \leq 5.2$ (63); the caustic-joint positions $n = 6501, 7506, 9193, 10615, 11259$ at $T = 13000$ against $n_f \approx T\sqrt{f}$; the drift arithmetic of §12.8 ($\mu(2) \Delta T \approx 360$ turns, 2264 rad; half/full-turn joints $n = 1041, 521$);

$\hat{d}_1 = \sqrt{7}d_1 = 0.2315$ at $T = 6.18$; the needle area $2 \times 1.03417 = 2.0683$; the values of $\overline{F}$ at half-integers; the κ_{1ps} identity once Change 5 is applied; and the bearing bookkeeping of §12.9 ($n_{\text{last}} = \lfloor t/\pi \rfloor \approx 2T^2$, $\nu = 1/4T^2$, arc-length ratio ≈ 236 at $t = 6.73 \times 10^8$).

Summary. 25 numbered changes: 8 errors, 1 standing inconsistency, 8 clarity items, 6 typo/citation items, 3 editorial (counting the optional ones). None has been applied to `main.tex`.